

UI/UX research and design spec: K!larity Enterprise Brain

Owner task: EB-81 Research: chat and answer UI kits (with EB-23 Web UI shell and EB-50 Ask Brain UI)
Date: 30 Sep 2026 · Status: first pass. Stack decided; the brand palette is provisional until the founder signs it off.

This page records what we studied, what we decided, and how every screen of the Brain should behave. The build that follows it is in `apps/web` (see §9).

1. What the UI has to do

The partners at Studio 8 will use the Brain to answer questions like "Why is Project Phoenix over budget?", often from a phone on site. They will act on the answers, which means spending money or chasing a client. So the UI has to make each answer **checkable in seconds**, not just readable.

Design principles

#	Principle	What it means on screen
P1	Evidence first	Every claim has a numbered citation. A citation opens the exact passage it relies on, not only the document.
P2	Numbers from the ledger	Figures show a "From ledger" badge naming the SQL query behind them (CLAUDE.md rule 3). The model never writes a number.
P3	Honest uncertainty	Every answer has a "What I couldn't confirm" section. With too little evidence, the Brain says so and does not guess.
P4	Nothing leaves without a person	Suggested actions only create drafts in Approvals (rule 10). The button says "Needs approval".
P5	Permission-aware, visibly	The composer says only accessible sources are searched. The nav and pages follow the user's role, and the server enforces it.
P6	One tenant on screen	Switching workspace resets every page. Synthetic or test tenants carry a badge.
P7	Calm and dense	Quiet neutrals and one action colour. Tabular numbers, Indian formats (₹18.4 lakh, IST) and mobile layouts that are first-class.
P8	Accessible by default	WCAG 2.1 AA contrast in light and dark. Full keyboard use, focus returned after dialogs close, reduced motion respected.

2. Tools and products studied

Current versions and licences were checked with `npm view` on 30 Sep 2026. "Pattern only" means we use the idea and copy no code or visual design.

Tool / product	Licence	Strong at	What we take	Verdict
assistant-ui (0.15.22)	MIT	Headless Thread/Composer primitives, tool-call UI, branch switching	Composer behaviour (Enter to send, Shift+Enter for a newline, IME-safe)	Not now. Our answer is a structured card, not a chat transcript. Revisit for multi-turn threads.
Vercel AI SDK (ai 7.0.x)	Apache-2.0	<code>streamObject</code> streams typed JSON; <code>useChat</code>	Stream the answer contract section by section	Adopt when EB-47 streams (server side). Needs rule-7 sign-off.
CopilotKit (1.75)	MIT	AG-UI protocol, shared agent state, human-in-the-loop interrupts	Approve/edit/reject interrupt pattern for Approvals	Pattern. Revisit with EB-66.
LibreChat	MIT	Conversation forking, artifacts, MCP tools	Fork-a-question idea for later	Pattern.
Open WebUI	BSD-3 plus branding clause	Citation popovers, knowledge-base RBAC	Nothing copied	Pattern only (branding clause).
Onyx web UI	MIT (ee excluded)	Inline numbered citations; a document sidebar splitting cited from other documents, with match highlights	Numbered chips, plus a side sheet that opens the source with the matched span highlighted	Pattern adopted. No code copied.
Perplexity	Proprietary	Ask → synthesized answer → inspect sources → dig further	A prominent numbered source list under each answer	Pattern.
Glean	Proprietary	Permission-aware enterprise search; the move from "ask" to "delegate"	Permission cue in the composer; delegation becomes Approvals	Pattern.
Hebbia Matrix	Proprietary, patented (US 12,393,788 "Matrix user interface for LLM-powered data analysis")	Documents × questions grid with cell-level citations	Nothing yet	Legal review before building any grid of prompts over documents (for example, comparing vendor quotations).
Apache Superset (embedded SDK)	Apache-2.0	Embedded dashboards with guest tokens and RLS	Possible later for Finance	Decided in EB-82.
Metabase	AGPL + commercial	Question builder, auto dashboards	Nothing	Pattern only.
Dify	Modified Apache-2.0	Workflow UI	Nothing. The licence bars copying its UI or interaction design.	Avoid.

Research findings that shaped the design

- **A wrong citation costs more than a missing one.** Studies show one valid citation earns as much trust as five, and an invalid citation erases the gain. So every citation must open the exact supporting span, and the UI never shows a citation it cannot open.

- **Show the work and the uncertainty.** Trust-focused AI UX guidance stresses confidence signals, visible reasoning and correction flows. That gives us the confidence badge with its reason, the "couldn't confirm" box, and feedback with reasons.
- **The shift toward delegation** (Glean and others): users increasingly assign goals, not questions. For us that becomes suggested actions that turn into approvable drafts, never direct side effects.

3. Decisions

ID	Decision	Why
UI-1	Next.js 16 (App Router) + React 19 + TypeScript 5.9. No component or CSS framework; design tokens in CSS custom properties; inline SVG icons.	Smallest licence surface (next, react, react-dom: MIT; TypeScript: Apache-2.0). Full control over accessibility. No Tailwind/shadcn/Radix to review yet.
UI-2	Pin Next 16.3.6 , not 16.3.7.	16.3.7 was one day old. pnpm's minimum-release-age guard flagged it, and we keep the guard on rather than add exclusions.
UI-3	The answer is a structured card , not free chat text. Sections: answer · key figures · why · risks · couldn't confirm · next steps · sources · feedback.	Mirrors the answer contract (EB-47); makes P1–P4 visible.
UI-4	Citations: numbered chips inline, plus a source list. A chip opens a side sheet with the passage and the supporting span highlighted, and what it was used to support.	P1; the Onyx and Perplexity patterns.
UI-5	Figures: a tile per figure with a "From ledger" badge; hovering shows the query name.	P2, rule 3.
UI-6	Insufficient evidence gets its own card state: no figures, no guesses, tips on how to rephrase.	P3.
UI-7	Actions create drafts for Approvals and show "Needs approval".	P4, rule 10.
UI-8	Tenant switch is a server action that resets to Ask. In production it is a re-login into the chosen Keycloak organization. The client never chooses the tenant for data access.	P6, ADR-013.
UI-9	Nav filtered by role, and pages guarded on the server (Finance and Executive: owner and partner; Settings: owner).	P5.
UI-10	Theme: light and dark from tokens. The preference is kept in a cookie so the server renders it with no flash.	P8.
UI-11	Streaming: for now, staged progress ("Understanding → Searching → Checking evidence"). Later, stream contract sections with the AI SDK.	Honest progress, without inventing a protocol before EB-47.
UI-12	Dev-only sign-in until Keycloak is ready. It is disabled in production builds, and the fixture answers are refused in production.	Unblocks UI work without weakening production.

4. Information architecture

Section	Page	Who	Purpose
Brain	Ask Brain (home)	everyone	Ask, verify, act
Brain	Projects	everyone	Per-project timeline, people, drawings, decisions, budget vs actual
Brain	Finance	owner, partner	Cash, receivables, budget variance, leakage flags
Brain	Decisions	everyone	Review draft decisions; confirmed decision log
Brain	Documents	everyone	Find the latest revision of any document or drawing
Workspace	Executive	owner, partner	State of the firm in under two minutes
Workspace	Approvals	everyone (approve rights by role)	Drafts waiting for a person
Workspace	Settings	owner	Sources, members, retention

The Kllarity operator console (tenants, health, impersonation) is a separate app, `apps/admin` (EB-88).

5. Screen specifications

Ask Brain. The empty state has a heading, the composer and four suggested questions. After a question, the answer thread uses a sticky composer. - While working: three staged steps and skeleton lines. - On error: an inline message with "Try again". - A 401 sends the user to sign-in. - Answers render as in UI-3.

Source sheet. A right-hand side sheet built on native `<dialog>`: Esc closes it, focus is trapped and returns to the citation. - **Header:** the source type, title, author, time in IST and project. - **Body:** the passage with the supporting span highlighted, and the claims it supports. - **Footer:** "Open in Gmail/WhatsApp/..." when the user may open the original.

Projects (EB-54). A list of projects with stage and health, then a project page containing: - a timeline; - people; - the latest drawings, with a revision badge; - open decisions; - budget vs actual, from SQL; - risks; - an Ask box scoped to the project.

Finance (EB-55). - Cash and receivables ageing. - Variance per project and package. - Overdue client payments. - Leakage flags.

Every number drills down to its ledger rows and evidence.

Decisions (EB-53). A review queue of drafts (confirm, edit, reject) with the source messages, plus a per-project decision log. Confirmed decisions answer "What did we decide..." first.

Documents (EB-57). Search with filters for project, type, revision and date, and a preview with a summary and extracted facts. Latest-revision badges on drawings.

Executive (EB-60). Tiles for cash, projects at risk, decisions waiting and overdue items, plus an "attention today" list. Every tile drills down to evidence.

Approvals (EB-66). Pending drafts showing: - who asked, and why; - the evidence behind the draft; - approve, edit or reject, each of which is audited.

Settings. Connected sources with sync health; members and roles; retention and export.

States every page must design for: - loading (skeleton); - empty (explain what will appear and why); - error (retry); - no access (explain; no data leaked); - insufficient evidence; - synthetic-tenant badge.

6. Visual language (provisional)

There is no K!larity brand guide yet, so these values are proposals for the founder to accept or replace. Change them only in `apps/web/styles/tokens.css`.

Token	Light	Dark	Use
<code>--brand</code>	#2B45D4	#8199FF	Actions, current nav item, links
<code>--accent</code>	#E89A0C	#F5B43A	The "!" in K!larity; the edge of highlighted evidence
<code>--mark-bg</code>	#FFEFC2	#4A3A12	Highlighted evidence span
<code>--text</code> / <code>--text-2</code> / <code>--text-3</code>	#0F172A / #445066 / #5F6B80	#E8ECF4 / #B0BACE / #95A0B8	Body, secondary, muted (all AA on their surfaces)
<code>--ok</code> / <code>--warn</code> / <code>--danger</code> / <code>--info</code>	#157A3A / #A4560A / #B42318 / #1F5FBF	lighter variants	Confidence, risk and status

- **Type:** the system UI font stack, so no web-font download. Sizes run from 12 to 28 px.
- **Radius:** 6, 10 and 14 px.
- **Layout:** sidebar 248 px, reading width 880 px, source sheet 440 px.

7. Component inventory

Component	Path	Status
App shell: sidebar, tenant switcher, top bar, mobile drawer	<code>components/shell/AppShell.tsx</code>	built
Theme toggle	<code>components/shell/ThemeToggle.tsx</code>	built
Icons (inline SVG)	<code>components/ui/Icon.tsx</code>	built
Page placeholder, no-access state	<code>components/ui/PagePlaceholder.tsx</code>	built
Ask view: composer, progress, thread	<code>components/answer/AskView.tsx</code>	built
Answer card	<code>components/answer/AnswerCard.tsx</code>	built
Citation list	<code>components/citations/CitationList.tsx</code>	built
Source sheet	<code>components/sources/SourcePanel.tsx</code>	built
Feedback	<code>components/answer/Feedback.tsx</code>	built (dev: acknowledged, not stored)
Data table, filters, timeline, KPI tile, chart	—	to build with EB-54, EB-55, EB-57 and EB-60
Command palette (⌘K)	—	proposed (new task)

8. Open questions and risks

1. **Brand:** accept or replace the provisional palette and logo tile.
2. **Answer contract:** `apps/web/lib/contracts.ts` is the UI's proposal. EB-47 owns the schema, and the TypeScript types should be generated from it.
3. **The Hebbia patent** limits any documents × questions grid feature. It needs legal review first.
4. **Keycloak:** sign-in, organization claim and tenant switch are dev stand-ins until the SSO task lands.
5. **Languages:** Telugu and Hindi site terms appear in sources. UI localisation is not planned yet.

9. What is built (30 Sep 2026)

- `apps/web` runs on Next 16.3.6. `pnpm install`, then `pnpm web:dev`, then open `http://localhost:3000`.
- In development, sign in as the demo user and pick a workspace.
- **Built and working:**
 - the shell (EB-23): login, role-filtered nav, tenant switch, light/dark, mobile drawer;
 - the Ask Brain flow (EB-50) on fixture data: answer card, citations, source sheet, "From ledger" figures, insufficient-evidence state, feedback, approval-gated actions.
- **Other Brain pages:** designed empty states with server-side role guards. Their data comes with their own tasks.
- **Checked** with a scripted browser walkthrough at desktop and 390 px mobile, in light and dark:
 - citation → exact span;
 - Esc closes, and focus returns to the citation;
 - a partner is refused Settings;
 - a tenant switch resets the view;
 - no horizontal overflow on mobile;
 - no console errors.
- `next build` and `tsc` pass.

10. Screens as built

Captured from the running app (fixture data, 30 Sep 2026).

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Sign in

Use your company account. You will only see answers from sources you have access to.

Continue with company SSO

Single sign-on (Keycloak) is not configured in this environment yet.

development only

Workspace

Studio 8 Hats (partner)

Continue as demo user

Sign in. SSO is shown but disabled until Keycloak lands; the demo sign-in appears in development only.

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S8 Studio 8 Hats

BRAIN

Ask Brain

Projects

₹ Finance

Decisions

Documents

WORKSPACE

Executive

Approvals

Permission-aware answers

Ask Brain

What do you want to know?

Ask anything about Studio 8 Hats. Answers come from your email, chats, documents and ledgers, with the source for every claim. Figures come from the ledger, never from the AI.

Ask about a project, decision, payment or document...

Only sources you can access are searched

Ask

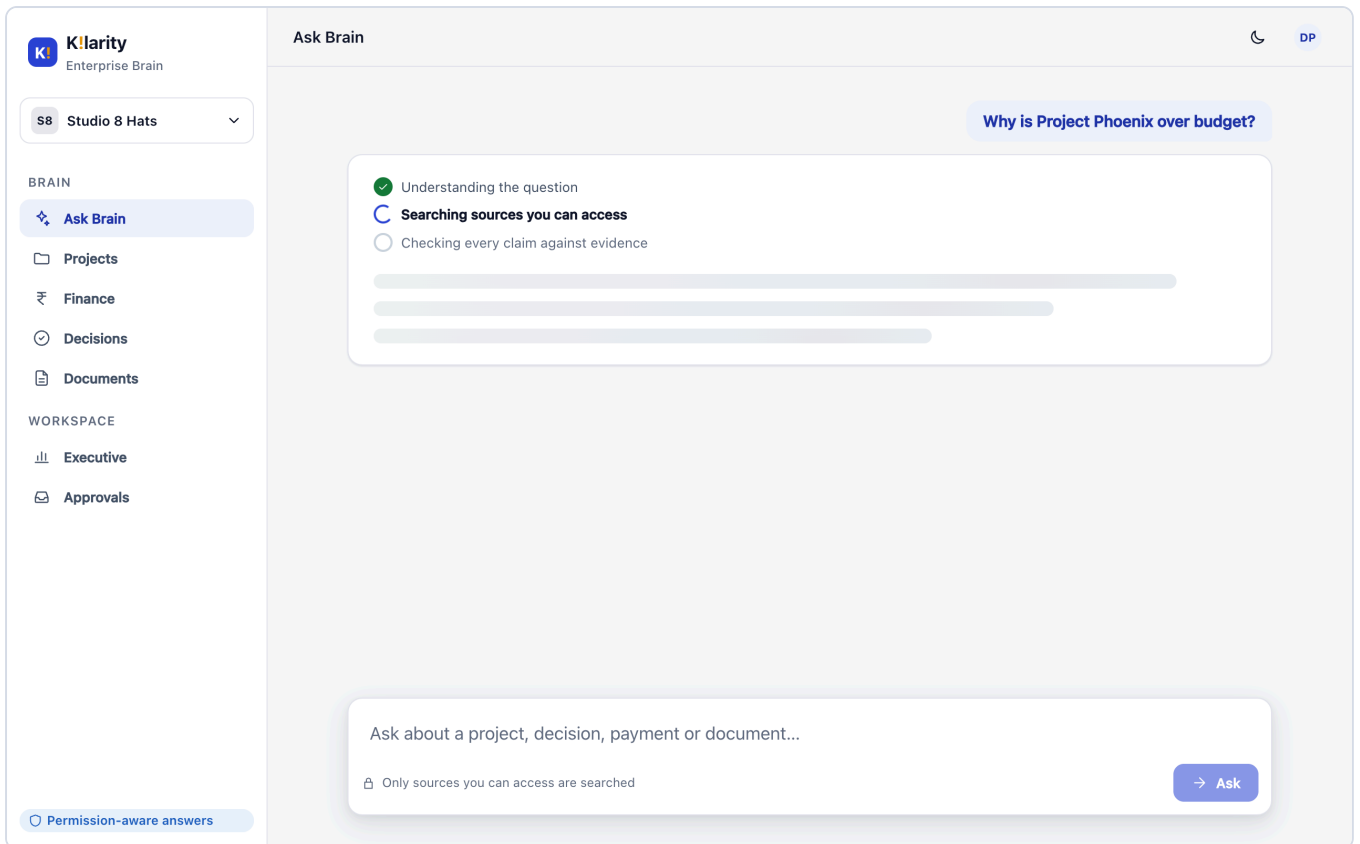
Why is Project Phoenix over budget?

What did we decide about the Phoenix facade?

What changed on Phoenix this week?

Which client payments are overdue?

Ask Brain, empty state: one question box, suggested questions, and a note that only accessible sources are searched.



While working: staged progress instead of a spinner.

High confidence 6 sources · All figures come from the finance ledger and every cause has a direct source.

Project Phoenix is ₹18.4 lakh (12%) over its approved budget. ¹ Half of that comes from the client's switch from ACP to HPL on the front facade, ² ³ a third from the steel transfer beam added in the Rev C structural drawings, ⁴ and the rest from labour kept on site while steel delivery slipped. ⁵ The facade change is not yet billable: its variation order is still unsigned. ⁶

Budget variance

₹18.4 lakh

 From ledger ¹

Facade (ACP → HPL)

₹9.2 lakh

 From ledger ¹ ³

Structural steel (Rev C)

₹6.1 lakh

 From ledger ¹ ⁴

Site labour (steel delay)

₹3.1 lakh

 From ledger ¹ ⁵

WHY

- Client approved HPL facade panels over ACP on 12 Aug. ² ³
- GFC Rev C replaced the RCC transfer beam with a steel section. ⁴
- Steel delivery slipped 3 weeks; crews were retained. ⁵

RISKS

- **high** ₹9.2 lakh facade variation is unrecoverable until VO-07 is signed. ⁶

WHAT I COULDN'T CONFIRM

- Whether the Rev C steel change can be billed to the client — no variation order or client message found.
- Labour cost after 30 Sep is not in the ledger yet.

SUGGESTED NEXT STEPS

Draft a reminder to the client to sign VO-07 Needs approval

Create a task: raise a variation for the Rev C steel Needs approval

SOURCES

- ¹  **Phoenix — Budget vs Actual (Finance ledger)**
Sheet · 28 Sept 2026, 6:30 pm
- ²  **Phoenix client group**
WhatsApp · Client representative · 12 Aug 2026, 11:04 am
- ³  **Revised quotation — facade cladding (HPL)**
Email · Facade vendor · 19 Aug 2026, 4:20 pm
- ⁴  **PHX-STR-204 Rev C (GFC) — transfer beam**
Drawing · Structural consultant · 30 Jul 2026, 10:00 am
- ⁵  **Site update — week 36**
Email · Site engineer · 6 Sept 2026, 7:12 pm
- ⁶  **Variation order VO-07 (draft, unsigned)**
Document · 2 Sept 2026, 12:00 pm

Was this answer useful?  **Yes**  **No**

The answer card: confidence with its reason, inline numbered citations, key figures marked From ledger, why, risks, what could not be confirmed, approval-gated next steps, sources and feedback.

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S8 Studio 8 Hats

BRAIN

Ask Brain

Projects

Finance

Decisions

Documents

WORKSPACE

Executive

Approvals

Permission-aware answers

Ask Brain

High confidence

6 sources - All figures come from the finance ledger and even

Project Phoenix is ₹18.4 lakh (12%) over its approved budget. ¹

from ACP to HPL on the front facade, ² ³ a third from the steel drawings, ⁴ and the rest from labour kept on site while steel delivery yet billable: its variation order is still unsigned. ⁶

Budget variance

₹18.4 lakh

From ledger ¹

Facade (ACP → HPL)

₹9.2 lakh

From ledger ¹ ³

Structure

₹6.1

From ledger ¹ ³

WHY

- Client approved HPL facade panels over ACP on 12 Aug. ² ³
- GFC Rev C replaced the RCC transfer beam with a steel section. ⁴
- Steel delivery slipped 3 weeks; crews were retained. ⁵

RISKS

- high** ₹9.2 lakh facade variation is unrecoverable until VO-07 is signed

Ask about a project, decision, payment or document...

Only sources you can access are searched

SUGGESTED NEXT STEPS

WhatsApp

Phoenix client group

From

Client representative

When

12 Aug 2026, 11:04 am IST

Project

Project Phoenix

Saw the HPL samples at site today. We prefer the HPL panels over ACP for the front facade even if it costs more — please go ahead and update the drawings.

The highlighted passage is what the answer relies on.

Used to support

- Client approved HPL facade panels over ACP on 12 Aug.

Open original

Clicking citation 2 opens the source with the exact supporting passage highlighted and the claim it supports.

Not enough evidence

I couldn't find enough in your connected sources to answer this reliably, so I won't guess.

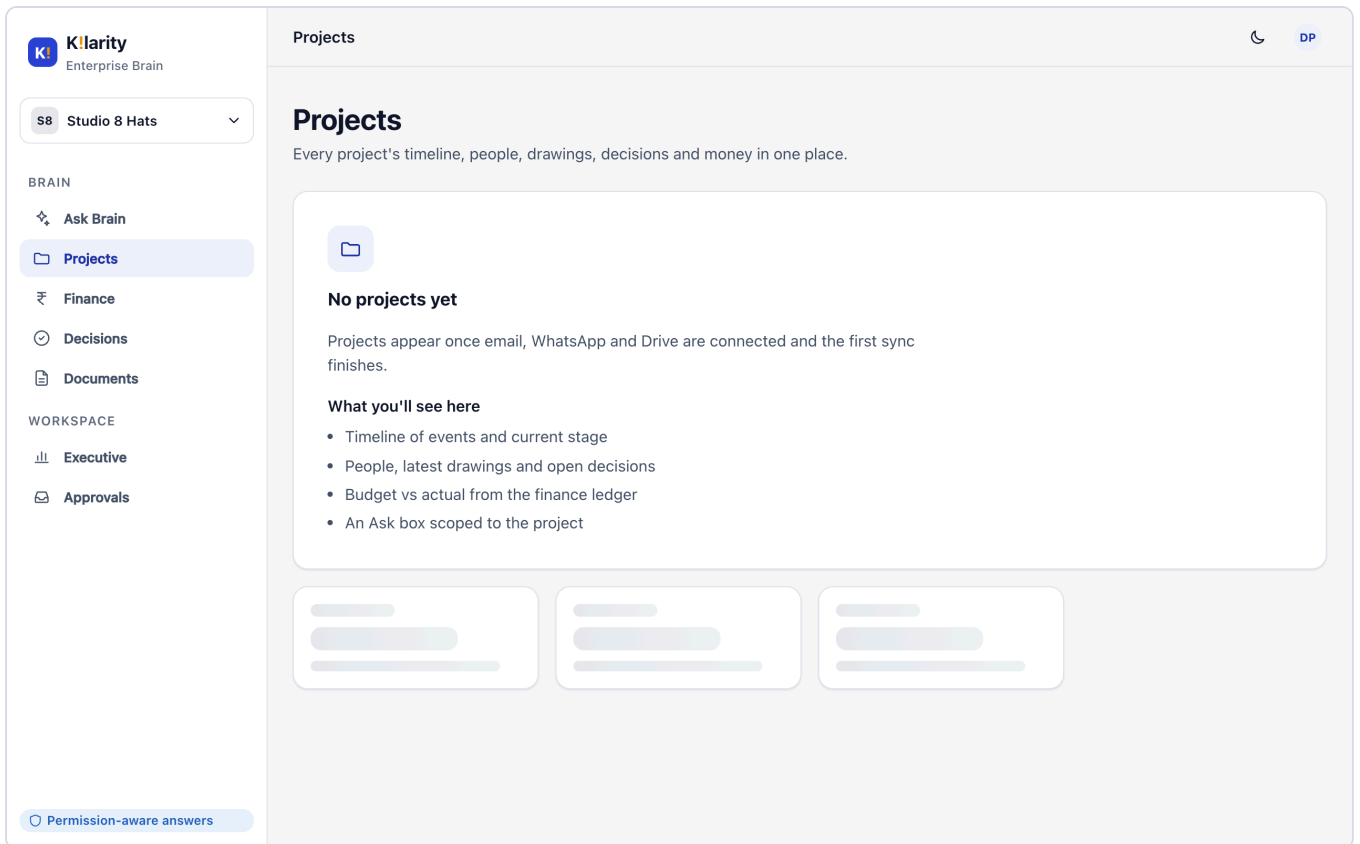
- No messages, emails, documents or ledger rows matched this question closely enough.
- Try naming the project, person or document, or ask an admin whether the right source is connected.

Was this answer useful?

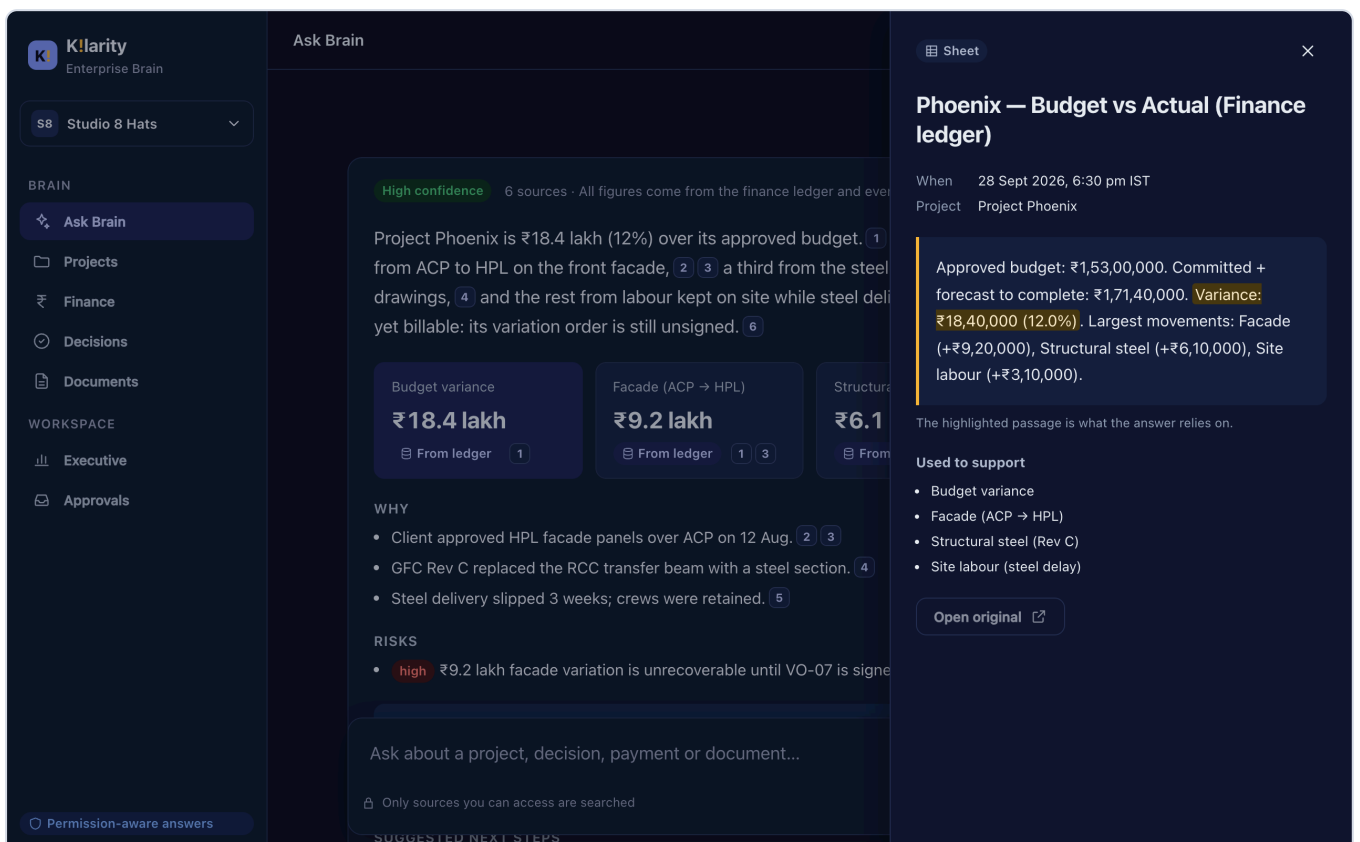
Yes

No

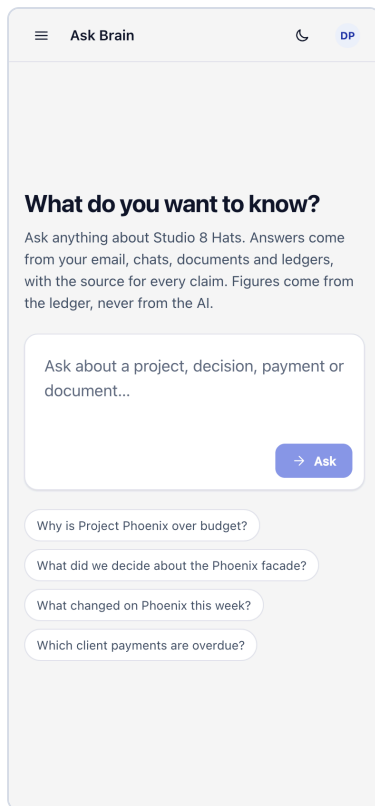
Not enough evidence: no guess, no figures, and tips on what to try.



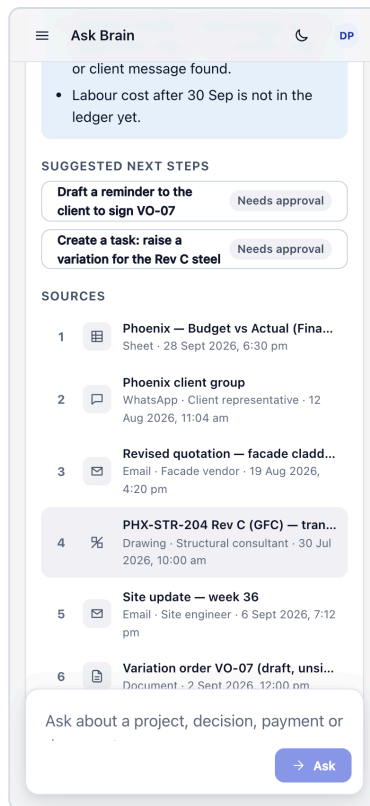
Other Brain pages have designed empty states until their data is connected.



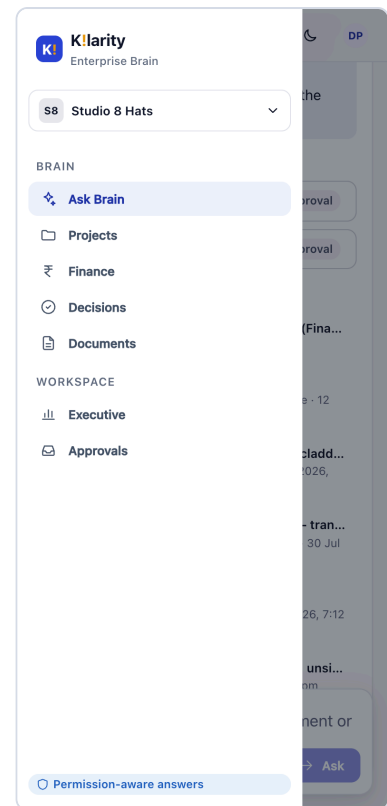
Dark theme: same layout, tokens swapped.



Mobile: ask



Mobile: answer



Mobile: navigation

Sources

- [AI UX patterns: the complete playbook for 2026 \(Lazarev\)](#)
- [UX patterns that increase trust in AI \(Goji Labs\)](#)
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- [Designing for confident wrong answers](#)
- [Perplexity search interface strategy \(Sacra\)](#)
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- [Hebbia \(Wikipedia\)](#) · [Matrix user interface patent US 12,393,788](#) · [Hebbia design interview](#)
- Onyx web UI read locally at commit a18fc1a: `web/src/sections/document-sidebar/` (pattern only, not copied)
- Tool Borrow Register (Notion) entries for assistant-ui, Vercel AI SDK, CopilotKit, LibreChat, Open WebUI, Superset, Metabase and Dify